



Yakeen Plus (2025)

Practice Test - 17

DURATION : 200 Minutes

DATE : 23/12/2024

M. MARKS : 720

Topics Covered

Physics :	Optics, Dual Nature of Matter and Radiation, Atoms and Nuclei, Electronic Devices
Chemistry :	Chemical Equilibrium, Ionic Equilibrium, Structure of Atom, Amines, Biomolecules, Practical Organic Chemistry, Salt Analysis
Biology :	(Botany): Molecular Basis of Inheritance, Organisms and Population, Ecosystem, Biodiversity and Conservation (Zoology): Human Health and Diseases, Biotechnology: Principles & Processes, Biotechnology and its Applications

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are **four Section** in the Question Paper, **Section I, II, III & IV** consisting of Section-I (Physics from question number **1 to 50**), Section-II (Chemistry: from question number **51 to 100**), Section-III (Botany: from question number **101 to 150**) & Section IV (Zoology: from question number **151 to 200**), having 50 Questions in each Subject and each subject is divided in two Section, Section A consisting 35 questions (all questions are compulsory) and Section B consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

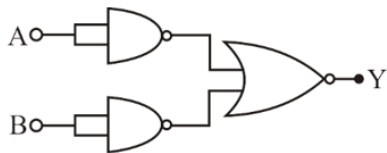
Practice Test-17

Yakeen Plus (2025)

Q1

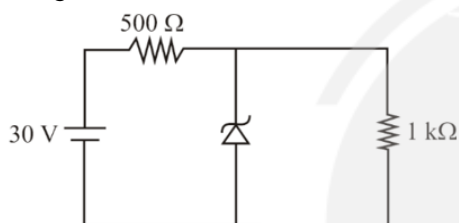
SECTION-A

The operation performed by the circuit as shown in figure, is;



- (1) OR (2) NOR
(3) AND (4) NAND

Q2 A zener diode of breakdown voltage $V_z = 10\text{ V}$ is used in the circuit given below. The current through the zener diode is:



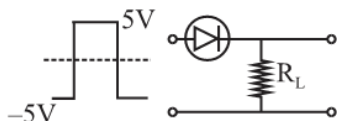
- (1) 40 mA (2) 30 mA
(3) 20 mA (4) 10 mA

Q3 The angular momentum of an electron in the hydrogen atom is $\frac{3h}{2\pi}$. Here, h is Planck's constant.

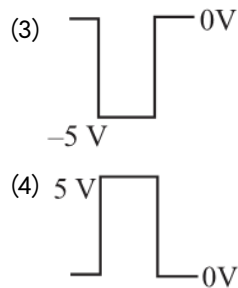
The kinetic energy of this electron is:

- (1) 4.35 eV (2) 1.51 eV
(3) 3.4 eV (4) 6.8 eV

Q4 If in a P-N junction diode a square input signal of 10 V is applied as shown, then the output signal across R_L will be:



- (1) 10 V
0 V
(2) -10 V
0 V



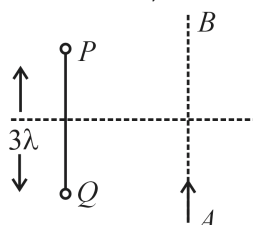
Q5 An unpolarised light is successively passed through two polaroids. Their transmission axis make 60° with each other. If the intensity of unpolarised light be I_0 , then the intensity after second polariser is

- (1) $\frac{I_0}{2}$
(2) $\frac{I_0}{4}$
(3) $\frac{I_0}{8}$
(4) $\frac{I_0}{3}$

Q6 In a double slit experiment, the distance between the slits is d . The screen is at a distance D from the slits. If bright fringe is formed opposite to one of the slits then its order is

- (1) $\frac{d}{\lambda}$
(2) $\frac{d^2}{2D\lambda}$
(3) $\frac{D^2}{2\lambda d}$
(4) $\frac{\lambda^2}{dD}$

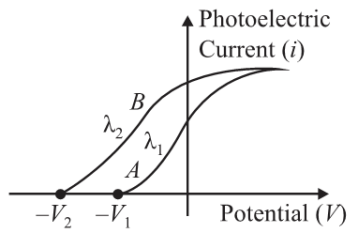
Q7 Two coherent light sources, each of wavelength λ , are separated by a distance 3λ . The maximum number of minima formed on line AB , which runs from $-\infty$ to $+\infty$, is



- (1) 2 (2) 4
(3) 6 (4) 8



- Q8** Select the **correct** statement for the following graph in a photoelectric experiment. Here λ_1, λ_2 are incident wavelengths, V_1, V_2 are stopping potentials.



- (1) maximum kinetic energy of photoelectrons is greater in curve B than curve A.
 (2) $\lambda_1 > \lambda_2$
 (3) frequency of incident radiation in curve B is greater than A.
 (4) all of these.

- Q9** Three light waves combine at a certain point, where their electric field components are $E_1 = E_0 \cos \omega t$, $E_2 = E_0 \sin(\omega t + 53^\circ)$, $E_3 = E_0 \sin(\omega t - 37^\circ)$. Find their resultant component $E(t)$ at that point.

- (1) $\frac{\sqrt{85}}{5} \sin\left(\omega t + \tan^{-1} \frac{6}{7}\right)$
 (2) $E_0 \sin(\omega t + 60^\circ)$
 (3) $E = \frac{\sqrt{85}}{5} \sin\left(\omega t + \tan^{-1} \frac{7}{6}\right)$

(4) None of these

- Q10** Two coherent point sources s_1 and s_2 vibrating in phase emit light of wavelength λ . The separation between the sources is 2λ . The smallest distance from s_2 on a line passing through s_2 and perpendicular to $s_1 s_2$, where a minimum of intensity occurs is

- (1) $\frac{7\lambda}{12}$
 (2) $\frac{15\lambda}{4}$
 (3) $\frac{\lambda}{2}$
 (4) $\frac{3\lambda}{4}$

- Q11** The two coherent sources of equal intensity produce maximum intensity of 100 units at a point. If the intensity of one of the source is reduce by 36% by reducing its width then the intensity of light at the same point will be:

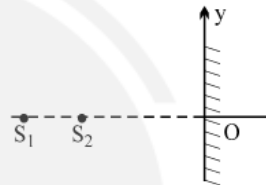
- (1) 90 (2) 89

- (3) 67 (4) 81

- Q12** Two monochromatic (wavelength = $a/5$) and coherent sources of electromagnetic waves are placed on the x -axis at the points $(2a, 0)$ and $(-a, 0)$. A detector moves in a circle of radius $R (> 2a)$ whose centre is at the origin. The number of maxima as detected during one circular revolution by the detector are:

- (1) 60 (2) 15
 (3) 64 (4) None of these

- Q13** Two point monochromatic and coherent sources of light of wavelength λ are placed on the dotted line in front of an large screen. The source emit waves in phase with each other. The distance between S_1 and S_2 is 'd' while their distance from the screen is much larger. Then, which of the following is/are **incorrect** statement(s)?



- (1) If $d = 7\lambda/2$, O will be a maxima
 (2) If $d = 4.3\lambda$, there will be a total of 8 minima on y axis.
 (3) If $d = 7\lambda$, O will be a maxima.
 (4) If $d = \lambda$, there will be only one maxima on the screen.

- Q14** In a Young's double slit experiment, the fringe width is found to be 0.4 mm . If the whole apparatus is immersed in water of refractive index $4/3$ without disturbing the geometrical arrangement, the new fringe width will be
- (1) 0.30 mm (2) 0.40 mm
 (3) 0.53 mm (4) 450 micron

- Q15** In young's double slit experiment the y-co-ordinates of central maxima and 10th maxima are 2 cm and 5 cm respectively. When the YDSE apparatus is immersed in a liquid of refractive index 1.5 the corresponding y-co-ordinates will be

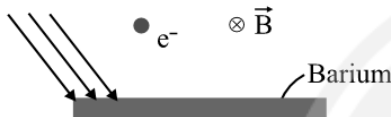
- (1) 2 cm, 7.5 cm
 (2) 3 cm, 6 cm
 (3) 6 cm, 3 cm
 (4) 2 cm, 4 cm



Q16 Magnetic field at the centre (at nucleus) of the hydrogen like atoms (atomic number = Z) due to the motion of electron in n th orbit is proportional to:

- (1) $\frac{n^3}{z^5}$
- (2) $\frac{n^4}{z}$
- (3) $\frac{z^2}{n^3}$
- (4) $\frac{z^3}{n^5}$

Q17 Light of wavelength $2475 \text{ \AA}$ is incident on Barium. Photoelectrons emitted describe a circle of radius 100 cm by a magnetic field of flux density $\frac{1}{\sqrt{17}} \times 10^{-5} \text{ Tesla}$. Work function of the Barium is (Given $e/m = 1.7 \times 10^{11}$)

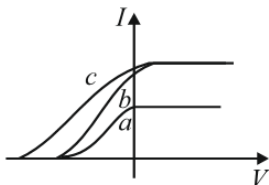


- (1) 1.8 eV
- (2) $21. \text{ eV}$
- (3) 4.5 eV
- (4) 3.3 eV

Q18 When a metallic surface is illuminated with radiation of wavelength λ , the stopping potential is V . If the same surface is illuminated with radiation of wavelength 2λ , the stopping potential is $\frac{V}{4}$. The threshold wavelength for the metallic surface is:

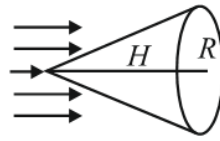
- (1) $\frac{V}{4}\lambda$
- (2) 3λ
- (3) 4λ
- (4) 5λ

Q19 The figure shows the variation of photocurrent with anode potential for a photosensitive surface for three different radiations. Let I_a , I_b and I_c be the intensities and f_a , f_b and f_c be the frequencies for the curves a , b and c respectively



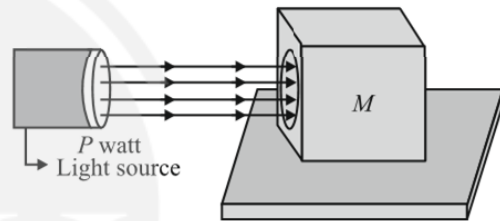
- (1) $f_a = f_b$ and $I_a \neq I_b$
- (2) $f_a = f_c$ and $I_a = I_c$
- (3) $f_a = f_b$ and $I_a = I_b$
- (4) $f_b = f_c$ and $I_b = I_c$

Q20 The radiation force experienced by a body exposed to radiation of intensity I assuming surface of body to be perfectly absorbing is:



- (1) $\frac{\pi R^2 I}{c}$
- (2) $\frac{2\pi R^2 I}{c}$
- (3) $\frac{4\pi R^2 I}{c}$
- (4) None of these

Q21 A source of light of power P is shown in Fig. Find the force on the block placed in the path of the light rays. The surface of body on which the light beam is incident is having a reflection coefficient $a_r = 0.7$ and absorption coefficient $a_a = 0.3$.



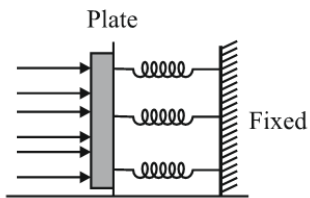
- (1) $F = 1.7 P/c$
- (2) $F = 2.7 P/c$
- (3) $F = 1.2 P/c$
- (4) $F = 1.6 P/c$

Q22 A small plane mirror (perfectly reflecting) of area A and mass M is hanging vertically with the help of a massless string of length l . Light of intensity I is incident horizontally on it. Find angle made by the string with the vertical.

- (1) $\theta = \tan^{-1} \frac{IA}{mgc}$
- (2) $\theta = \sin^{-1} \frac{I2A}{mgc}$
- (3) $\theta = \sin^{-1} \frac{IA}{mgc}$
- (4) $\theta = \tan^{-1} \frac{2IA}{Mgc}$

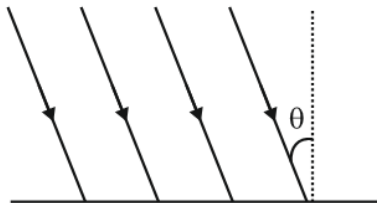
Q23 Light of intensity I is incident normally on a perfectly reflecting plate of area A kept in a gravity free space. If the photons strike the plate symmetrically and initially the spring was at its natural length, find the maximum compression in the springs





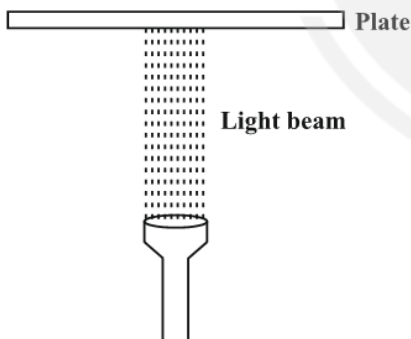
- (1) IA/KC (2) $2IA/3KC$
 (3) $3IA/KC$ (4) $4IA/3KC$

Q24 Calculate force exerted by a light beam on a surface if light of Power P is incident on the surface at an angle θ as shown. Consider surface to be perfectly reflecting.



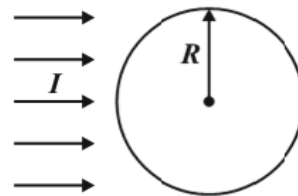
- (1) $P \cos \theta / c$
 (2) $2P \sin \theta / c$
 (3) $2P \cos \theta / c$
 (4) None of these

Q25 A plate of mass 10 g is in equilibrium in air due to the force exerted by light beam on plate. Calculate power of beam. Assume plate is perfectly absorbing.



- (1) $1.5 \times 10^7 \text{ W}$
 (2) $3 \times 10^7 \text{ W}$
 (3) $4.5 \times 10^7 \text{ W}$
 (4) $6 \times 10^7 \text{ W}$

Q26 Consider a sphere of radius R exposed to radiation of intensity I as shown in figure. If surface of sphere is partially reflection and reflection coefficient is 0.3, then radiation force experienced is



- (1) $\frac{\pi R^2 I}{c}$
 (2) $\frac{1.7\pi R^2 I}{c}$
 (3) $\frac{1.3\pi R^2 I}{c}$
 (4) None of these

Q27 Calculate the binding energy of a $^{120}_{50}\text{Sn}$ nucleus. Given atomic mass of $\text{Sn}^{120} = 119.902199 \text{ amu}$, mass of proton, $m_p = 1.007825 \text{ amu}$, mass of neutron, $m_n = 1.008665 \text{ amu}$.

- (1) 102.38 MeV (2) 1020.38 MeV
 (3) 931.5 MeV (4) None of these

Q28 Calculate the binding energy when two neutron and two protons combine to form an α -particle. Given that mass of neutron = 1.00893 amu, mass of proton = 1.00813 amu, mass of deuteron = 2.01473 amu and mass of α -particle = 4.00389 amu

- (1) 281 MeV (2) 2.81 MeV
 (3) 3.71 MeV (4) 28.1 MeV

Q29 Consider the following nuclear reaction
 $X^{200} \rightarrow A^{110} + B^{90} + \text{Energy}$

If the binding energy per nucleon for X , A and B are 7.4 MeV, 8.2 MeV and 8.2 MeV respectively, the energy released will be:

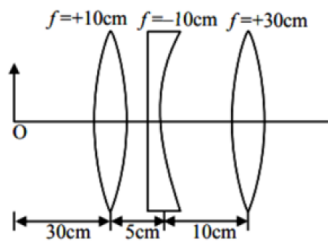
- (1) 90 MeV (2) 110 MeV
 (3) 200 MeV (4) 160 MeV

Q30 In the reaction $^2_1\text{H} + ^3_1\text{H} \rightarrow ^4_2\text{He} + ^1_0\text{n}$. If the binding energies of ^2_1H , ^3_1H and ^4_2He are respectively a , b and c (in MeV), then the energy (in MeV) released in this reaction is:

- (1) $a + b + c$ (2) $c + a - b$
 (3) $c - a - b$ (4) $a + b - c$

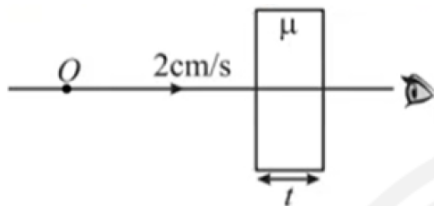
Q31 Find the distance of the image from object O , formed by the combination of lenses in the figure:





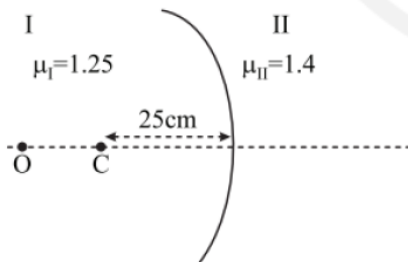
- (1) 75 cm (2) 20 cm
(3) 10 cm (4) infinity

Q32 A glass slab of width t , refractive index μ is placed as shown in the figure. If the point object, moves with a speed 2 cm/s towards the slab the speed observed will be



- (1) 2 cm/s
(2) Less than 2 cm/s
(3) Greater than 2 cm/s
(4) Dependent on the refractive index of surrounding medium

Q33 Region I and II are separated by a spherical surface of radius 25 cm . An object is kept in region I at a distance of 40 cm from the surface. The distance of the image from the surface is:

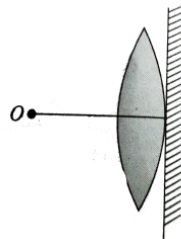


- (1) 9.52 cm (2) 18.23 cm
(3) 37.58 cm (4) 55.44 cm

Q34 A ray of light from a denser medium strike a rarer medium. The angle of reflection is r and that of refraction is r' . The reflected and refracted rays make an angle of 90° with each other. The critical angle will be

- (1) $\sin^{-1}(\tan r)$
(2) $\tan^{-1}(\sin r)$
(3) $\sin^{-1}(\tan r')$
(4) $\tan^{-1}(\sin r')$

Q35 Behind a thin converging lens having both the surfaces of the same radius 10 cm , a plane mirror has been placed. The image of an object at a distance of 40 cm from the lens is formed at the same position. What is the refractive index of the lens?

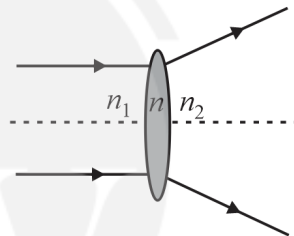


- (1) 1.5 (2) $5/3$
(3) $9/8$ (4) None of these

Q36

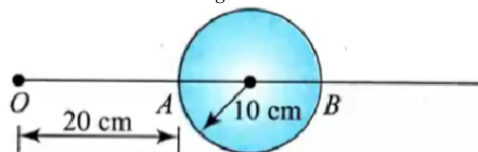
SECTION-B

If the behavior of light rays is as shown in the figure given below. What is the correct relation between the refractive indices n , n_1 and n_2 .



- (1) $n_2 < n$ (2) $n_2 > n$
(3) $n_2 = n$ (4) None of these

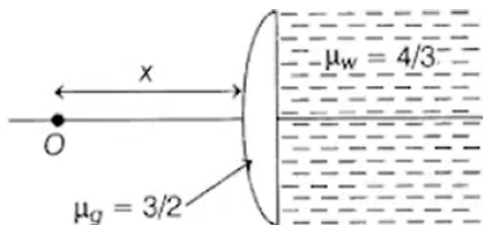
Q37 A glass sphere of radius $R = 10 \text{ cm}$ is kept inside water. A point object O is placed at 20 cm from A as shown in the figure. Find the position and nature of the image when seen from other side of the sphere. Given $\mu_g = 3/2$ and $\mu_w = 4/3$.



- (1) 200 cm , virtual
(2) 100 cm , real
(3) 100 cm , virtual
(4) 300 cm , virtual

Q38 An object O is kept in air in front of a thin plano-convex lens of radius of curvature 10 cm . Its refractive index is $3/2$ and the medium towards right of the plane surface is water of refractive index $4/3$. What should be distance x of the object so that the rays become parallel finally?



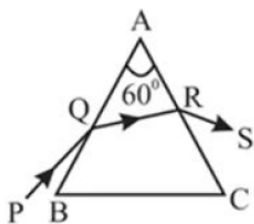


- (1) 5 cm (2) 10 cm
(3) 20 cm (4) 15 cm

Q39 The angle of minimum deviation measured with a prism is 30° and the angle of prism is 60° . The refractive index of prism material is:

- (1) $\sqrt{2}$ (2) 2
(3) $3/2$ (4) $4/3$

Q40 A ray PQ incident on the refracting face BA is refracted in the prism BAC as shown in the figure and emerges from the other refracting face AC as RS , such that $AQ = AR$. If the angle of prism $A = 60^\circ$ and the refractive index of the material of prism is $\sqrt{3}$, then the angle of deviation of the ray is:

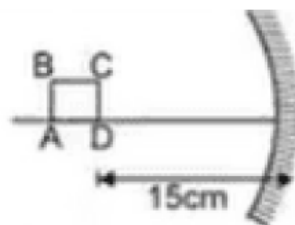


- (1) 60° (2) 45°
(3) 30° (4) None of these

Q41 A point object is moving with velocity $\vec{v}_o = 2\hat{i} - 3\hat{j} + 4\hat{k}$ in front of a moving plane mirror whose normal is along x -axis. The mirror is moving with velocity $\vec{v}_m = \hat{i} - 4\hat{j} + 3\hat{k}$. Find the velocity vector of the image:

- (1) $5\hat{j}$
(2) $-3\hat{j} + 4\hat{k}$
(3) $-4\hat{j} + 2\hat{k}$
(4) $2\hat{i} - 3\hat{j} + 2\hat{k}$

Q42 A square $ABCD$ of side 1 mm is kept at distance 15 cm in front of the concave mirror as shown in the figure. The focal length of the mirror is 10 cm. The length of the perimeter of its image will be (nearly):



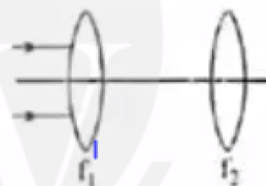
- (1) 8 mm (2) 2 mm
(3) 12 mm (4) 6 mm

Q43 A fish rising vertically up towards the surface of water with speed of 3 cm/s observes a bird diving vertically down towards it with a speed of 9 cm/s. Actual velocity of the bird is:

$$\left(\mu_w = \frac{4}{3}\right)$$

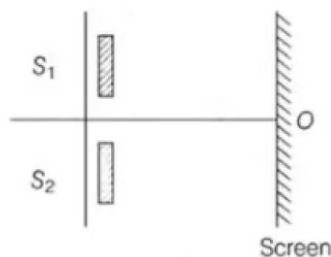
- (1) 4.5 cm/s (2) 5 m/s
(3) 3 m/s (4) 3.4 cm/s

Q44 Parallel beam of light is incident on a system of two convex lenses of focal lengths $f_1 = 20$ cm and $f_2 = 10$ cm. What should be the distance between the two lenses so that rays after refraction from both the lenses pass undeviated



- (1) 60 cm (2) 30 cm
(3) 90 cm (4) 40 cm

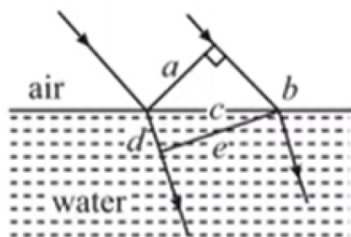
Q45 YDSE is carried with two thin sheets of thickness $10.4\mu\text{m}$ each and refractive index $\mu_1 = 1.52$ and $\mu_2 = 1.40$ covering the slits S_1 and S_2 , respectively. If white light of range 400 nm to 780 nm is used, then which wavelength will form maxima exactly at point O , the centre of the screen



- (1) 416 nm only
(2) 624 nm only
(3) 416 nm and 624 nm only
(4) None of these



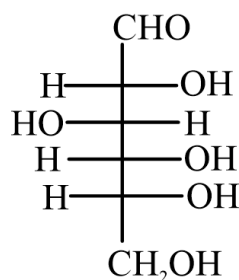
- Q46** Figure shown plane waves refracted from air to water using Huygens principle a, b, c, d, e are lengths on the diagram. The refractive index of water with respect to air is



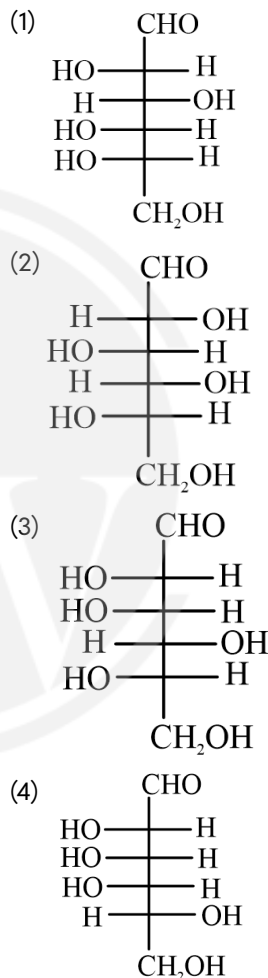
- (1) a/e (2) b/e
(3) b/d (4) d/b
- Q47** In a double-slit experiment, at a certain point on the screen the path difference between the two interfering waves is $\frac{1}{8}$ th of a wavelength. The ratio of the intensity of light at that point to that at the centre of a bright fringe is:
- (1) 0.568 (2) 0.760
(3) 0.853 (4) 0.672
- Q48** In a double slit experiment, when a thin film of thickness t having refractive index μ is introduced in front of one of the slits, the maximum at the centre of the fringe pattern shifts by one fringe width. The value of t is (λ is the wavelength of the light used).
- (1) $\frac{\lambda}{2(\mu-1)}$
(2) $\frac{\lambda}{(\mu-1)}$
(3) $\frac{\lambda}{(2\mu-1)}$
(4) $\frac{2\lambda}{(\mu-1)}$
- Q49** In YDSE, $d = 2$ mm, $D = 2$ m, and $\lambda = 500$ nm. If intensities of two slits are I_0 and $9I_0$, then find intensity at $y = \frac{1}{6}$ mm.
- (1) $7I_0$ (2) $10I_0$
(3) $16I_0$ (4) $4I_0$
- Q50** The maximum kinetic of a photoelectron is 3 eV. What is its stopping potential?
- (1) 3 V (2) 5 V
(3) 2 V (4) 1 V
- Q51** The limiting line in Balmer series will have a frequency of :
- (1) $6.22 \times 10^{15} \text{ s}^{-1}$
(2) $7.22 \times 10^{14} \text{ s}^{-1}$
(3) $8.22 \times 10^{14} \text{ s}^{-1}$

(4) $9.22 \times 10^{14} \text{ s}^{-1}$

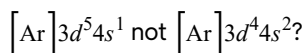
- Q52** The structure of D - (+)-glucose is:



The structure of L - (-)-glucose is:



- Q53** Why electronic configuration of Cr_{24} is



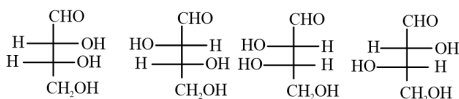
- (1) Energy of $3d$ orbital is more than $4s$ -orbital.
(2) According to Pauli's principle $[\text{Ar}] 3d^5 4s^1$ is acceptable.
(3) After shifting of 1 electron from $4s$ to $3d$ distribution of electron will be more symmetrical which impacts extra stability.
(4) All of these.



Q54 The vapour density of N_2O_4 at certain temperature is 30. What is the percentage dissociation of N_2O_4 at this temperature?

- (1) 53.3% (2) 106.6%
(3) 26.7% (4) 34.5%

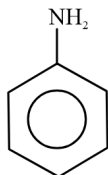
Q55 The **correct** corresponding order of names of four aldoses with configuration given below;



respectively, is:

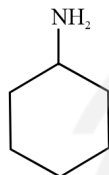
- (1) L-erythrose, L-threose, D-erythrose, D-threose
(2) D-threose, D-erythrose, L-threose, L-erythrose
(3) D-erythrose, L-threose, L-erythrose, D-threose
(4) D-erythrose, D-threose, L-erythrose, L-threose.

Q56



(aniline)

and



(cyclohexyl amine)

can be

differentiated by :

- (1) Hinsberg test
(2) Iso-cyanide test
(3) $NaNO_2/HCl, 0^\circ C$ then α -Naphthol
(4) NaOH

Q57 PCl_5 is in equilibrium with PCl_3 and Cl_2 in a cylinder provided with a piston. When the system is compressed with the help of the piston, which of following will happen?

- (1) Some more PCl_5 will decompose to give PCl_3 and Cl_2 .
(2) Some PCl_3 and Cl_2 will give more of PCl_5 .
(3) There will be no effect.
(4) Both (2) and (3)

Q58 An electron has Kinetic-energy (K.E) is $6.0 \times 10^{-25} J$, its wavelength will be:

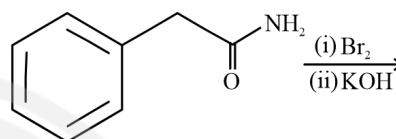
- (1) 8967 Å
(2) 5768 Å
(3) 6342 Å
(4) 4321 Å

Q59 A solution of m-chloroaniline, m-chlorophenol and m-chlorobenzoic acid in ethyl acetate was extracted initially with a saturated solution of $NaHCO_3$ to give fraction A. The left over organic

phase was extracted with dilute NaOH solution to give fraction B. The final organic layer was labelled as fraction C. Fractions A, B and C, contain respectively :

- (1) m-Chlorobenzoic acid, m-Chloroaniline and m-Chlorophenol
(2) m-Chloroaniline, m-Chlorobenzoic acid and m-Chlorophenol
(3) m-Chlorobenzoic acid, m-Chlorophenol and m-Chloroaniline
(4) m-Chlorophenol, m-Chlorobenzoic acid and m-Chloroaniline

Q60 When the following amide is treated with Br_2/KOH , it gives:



- (1)
- (2)
- (3)
- (4)

Q61 Kjeldahl's method **cannot** be used to estimate nitrogen for which of the following compound?

- (1) $C_6H_5NO_2$
(2) $C_6H_5NH_2$
(3) $CH_3CH_2-C \equiv N$
(4)

Q62 Some statements are given below:

1. Glucose is aldohexose.
2. Naturally occurring glucose is dextrorotatory.
3. Glucose contains three chiral centres.

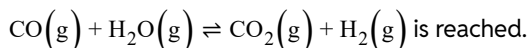


4. Glucose contains one 1° — alcoholic group and four 2° — alcoholic groups.

Among the above **correct** statement(s) are

- (1) 1 and 2
(2) 3 and 4
(3) 1, 2 and 4
(4) all are correct.

Q63 60 g of water gas and 18 g of steam are heated in a closed vessel to a temperature of 450°C so that the equilibrium ;



If dissociation Constant (K_a) for the reaction is 3 , the mass of CO_2 present will be:

- (1) 11 g (2) 22 g
(3) 44 g (4) 60 g

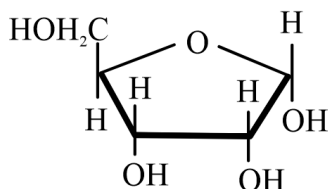
Q64 In which of the following reaction cyanide will be obtained as a major product?

- (1) $\text{Ph}-\text{C}(=\text{O})-\text{CH}_3 \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4}$
(2) $\text{Ph}-\text{C}(=\text{O})-\text{NH}_2 \xrightarrow{\text{Br}_2, \text{NaOH}}$
(3) $\text{Ph}-\text{C}(=\text{O})-\text{NH}_2 \xrightarrow{\text{P}_4\text{O}_{10}}$
(4) $\text{Ph}-\text{C}(=\text{O})-\text{OH} \xrightarrow[\text{NH}_3]{\text{SOCl}_2}$

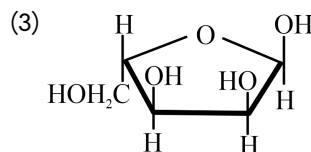
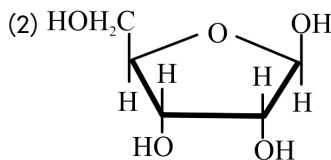
Q65 The amine which reacts with NaNO_2 and dil. HCl to give yellow oily compound is:

- (1) ethylamine (2) isopropylamine
(3) sec-butylamine (4) dimethylamine.

Q66 Which of the following represents the anomer of the compound shown?

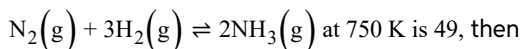


- (1)

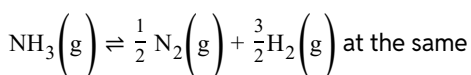


(4) None of these.

Q67 If the equilibrium constant for the reaction



at 750 K is 49, then the equilibrium constant for the reaction



at the same temperature is:

- (1) $1/49$ (2) 17
(3) 21 (4) $1/7$

Q68 Nessler's Reagent is used to detect

- (1) CrO_4^{2-} (2) PO_4^{3-}
(3) MnO_4^- (4) NH_4^+

Q69 CO_2 gas & SO_2 gas are distinguished by using

- (1) NaOH (2) NH_3
(3) $\text{K}_2\text{Cr}_2\text{O}_7$ (4) HCl

Q70 Which of the following compounds on treatment with NaNO_2/HCl at 0°C and then coupled with phenol produces p-hydroxyazobenzene?

- (1) Nitrobenzene
(2) Azobenzene
(3) Aniline
(4) Phenyl isocyanide

Q71 For a reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ at 721 K, the value of equilibrium constant is 50. If 0.5 mole each of H_2 and I_2 is added to the system the value of equilibrium constant will be:

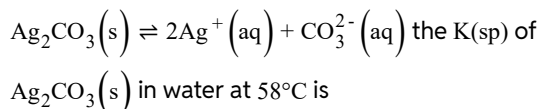
- (1) 0.02 (2) 0.2
(3) 50 (4) 25

Q72 Diabetes is detected by using _____ for testing urine of patients.

- (1) Fehling's solution
(2) Tollens' reagent
(3) Benedict's solution
(4) Baeyer's reagent



Q73 Using the Gibbs' energy change, $\Delta G^\circ = +63.3 \text{ kJ}$, for the following reaction,



$$(R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1})$$

- (1) 10^{-12} (2) 10^{-10}
(3) 10^{-14} (4) 10^{-16}

Q74 Prussian Blue is formed when:

- (1) Ferrous Sulphate reacts with FeCl_3
(2) Ferric Sulphate reacts with $\text{K}_4[\text{Fe}(\text{CN})_6]$
(3) Ammonium sulphate reacts with FeCl_3
(4) Ferrous Ammonium sulphate reacts with FeCl_3

Q75 Find the hydrolysis product of maltose.

- (1) α -D-glucose + α -D-glucose
(2) α -D-glucose + α -D-fructose
(3) α -D-glucose + α -D-galactose
(4) α -D-fructose + α -D-galactose

Q76 K_c for the reaction: $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ is 4.0×10^{-6} at 400 K. K_p for the above reaction is:

- (1) 2.4×10^{-3}
(2) 4.0×10^{-6}
(3) 4.0×10^{-8}
(4) 6.0×10^{-4}

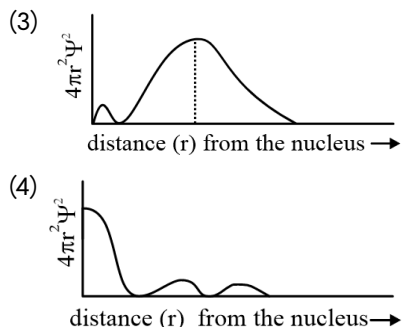
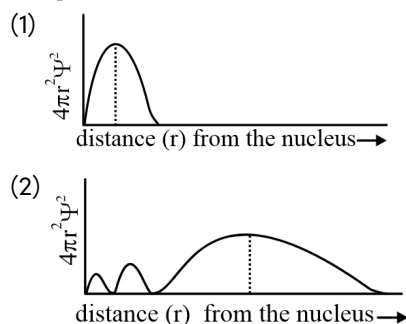
Q77 Which of the following gives black ppt. with NH_4OH ?

- (1) Ag_2S (2) AgCl
(3) HgCl_2 (4) Hg_2Cl_2

Q78 0.189 g of an organic substance gave in a carius determination 0.287 g of silver chloride. What is the % of Cl in the given compound?

- (1) 37.57% (2) 27.57%
(3) 73.57% (4) 47.57%

Q79 Which of the following is radial distribution curve for 3p-orbital?



Q80 The order of basic strength among the following amines in benzene solution is:

- (1) $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH}$
(2) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2$
(3) $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$
(4) $(\text{CH}_3)_3\text{N} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH}$

Q81 In a multi-electron atom, which of the following orbitals, described by the three quantum numbers, will have the same energy in the absence of magnetic field and electric field?

- (i) $n = 1, l = 0, m = 0$
(ii) $n = 2, l = 0, m = 0$
(iii) $n = 3, l = 1, m = 1$
(iv) $n = 3, l = 2, m = 1$
(v) $n = 3, l = 2, m = 0$
(1) (i) and (ii) (2) (ii) and (iii)
(3) (iii) and (iv) (4) (iv) and (v)

Q82 Accumulation of lactic acid ($\text{C}_3\text{H}_6\text{O}_3$), a monobasic acid in tissues leads to pain and a feeling of fatigue. In a 0.10 M aqueous solution, lactic acid is 3.7% dissociated. The value of dissociation constant, K_a , for this acid will be:

- (1) 1.4×10^{-5} (2) 1.4×10^{-4}
(3) 3.7×10^{-4} (4) 2.8×10^{-4}

Q83 $\text{HSO}_4^- + \text{OH}^- \rightarrow \text{SO}_4^{2-} + \text{H}_2\text{O}$. Which is correct about conjugate acid-base pair?

- (1) SO_4^{2-} is conjugate acid of base HSO_4^-
(2) HSO_4^- is conjugate base of acid SO_4^{2-}
(3) SO_4^{2-} is conjugate base of acid HSO_4^-
(4) none of these.

Q84 Ionisation constant (K_a) of CH_3COOH is 1.7×10^{-5} and concentration of H^+ ions is $3.4 \times 10^{-4} \text{ M}$. Then find out initial concentration of CH_3COOH .



- (1) $3.4 \times 10^{-4} \text{M}$
 (2) $3.4 \times 10^{-3} \text{M}$
 (3) $6.8 \times 10^{-4} \text{M}$
 (4) $6.8 \times 10^{-3} \text{M}$

Q85 The amine which can react with $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ can be:

- (1) primary amine only
 (2) secondary amine only
 (3) tertiary amine only
 (4) both primary and secondary amines

Q86 In the Bohr's orbit, what is the ratio of total kinetic energy and total energy of the electron?

- (1) -1 (2) -2
 (3) +1 (4) +2

Q87 The pH of 0.004 M hydrazine solution is 9.7. Its ionisation constant (K_b) is:

(Given: $\text{antilog}(-4.3) = 5 \times 10^{-5}$)

- (1) 7.79×10^{-8}
 (2) 4.49×10^{-9}
 (3) 1.67×10^{-10}
 (4) 6.25×10^{-7}

Q88 A golf ball has a mass of 40 g and a speed of 45 m/s. If the speed can be measured within accuracy of 2%. The uncertainty in the position is

- (1) $2.45 \times 10^{-30} \text{ m}$
 (2) $2.78 \times 10^{-33} \text{ m}$
 (3) $1.46 \times 10^{-33} \text{ m}$
 (4) $4.75 \times 10^{-20} \text{ m}$

Q89 What is the pH of 0.01 M glycine solution?

For glycine $K_{a1} = 4.5 \times 10^{-3}$ and

$K_{a2} = 1.7 \times 10^{-10}$ at 298 K.

- (1) 3.0 (2) 10.0
 (3) 7.06 (4) 8.2

Q90 What is the formula of Glassy bead?

- (1) NaPO_3 (2) NaBO_2
 (3) $\text{NaBO}_2 + \text{B}_2\text{O}_3$ (4) None of these

Q91 Aniline when acetylated, the product on nitration followed by alkaline hydrolysis gives:

- (1) acetanilide
 (2) o-nitroacetanilide
 (3) p-nitroaniline
 (4) m-nitroaniline

Q92 The pK_a of formic acid and pK_b of ammonium hydroxide are 3.75 and 4.75 respectively.

Calculate the pH of ammonium formate solution.

- (1) 5.01 (2) 0.45
 (3) 9.006 (4) 6.5

Q93 Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion : The two strands of DNA are complementary to each other.

Reason : Adenine specifically forms hydrogen bonds with guanine whereas cytosine forms hydrogen bonds with thymine.

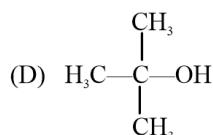
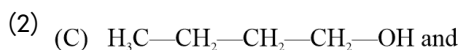
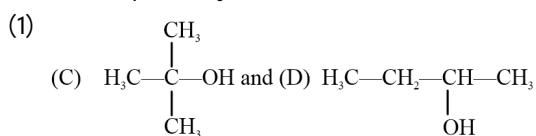
In the light of the above statements, choose the **correct** answer from the options given below:

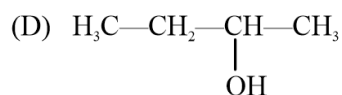
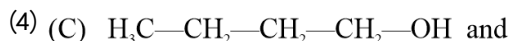
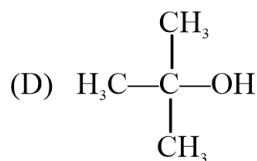
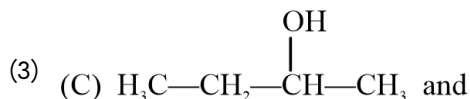
- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.

Q94 Two compounds A and B with same molecular formula ($\text{C}_3\text{H}_6\text{O}$) undergo Grignard's reaction with methylmagnesium bromide to give products C and D. Products C and D show following chemical tests.

Test	C	D
Ceric ammonium nitrate Test	Positive	Positive
Lucas Test	Turbidity obtained after five minutes	Turbidity obtained immediately
Iodoform Test	Positive	Negative

C and D respectively are :





Q95 The $\text{p}K_a$ of HCN is 9.30. The pH of a solution prepared by mixing 2.5 moles of KCN and 2.5 moles of HCN in water and making the total volume upto 500 mL is:

- (1) 9.30 (2) 7.30
(3) 10.30 (4) 8.30

Q96 Match the following :

List - I (Test/Method)	List - II (Reagent)
(i) Lucas Test	(A) $\text{C}_6\text{H}_5\text{SO}_2 \text{ Cl/KOH}$
(ii) Dumas method	(B) CuO/CO_2
(iii) Kjeldahl's method	(C) Conc. HCl and ZnCl_2
(iv) Hinsberg Test	(D) H_2SO_4

- (1) (i)-(C), (ii)-(B), (iii)-(D), (iv)-(A)
(2) (i)-(B), (ii)-(D), (iii)-(C), (iv)-(A)
(3) (i)-(D), (ii)-(C), (iii)-(B), (iv)-(A)
(4) (i)-(B), (ii)-(A), (iii)-(C), (iv)-(D)

Q97 $A(g) \rightleftharpoons B(g) + \frac{1}{2}C(g)$ The correct relationship

between K_p , α and equilibrium pressure P is

- (1) $K_p = \frac{\alpha^{1/2}P^{3/2}}{(2+\alpha)^{3/2}}$
(2) $K_p = \frac{\alpha^{3/2}P^{1/2}}{(2+\alpha)^{1/2}(1-\alpha)}$
(3) $K_p = \frac{\alpha^{1/2}P^{1/2}}{(2+\alpha)^{3/2}}$
(4) $K_p = \frac{\alpha^{1/2}P^{1/2}}{(2+\alpha)^{1/2}}$

Q98 Match the List I with column List II, and select the correct option.

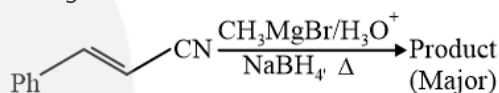
List - I	List - II
(i) Acidic amino acid	(A) Glycine
(ii) Sulphur containing amino acid	(B) Histidine
(iii) Essential amino acid	(C) Cysteine
(iv) Optically inactive amino acid	(D) Glutamic acid

- (1) i-(A), ii-(B), iii-(C), iv-(D)
(2) i-(D), ii-(B), iii-(A), iv-(C)
(3) i-(B), ii-(D), iii-(A), iv-(C)
(4) i-(D), ii-(C), iii-(B), iv-(A)

Q99 Number of photons emitted in 5 hours by a 50 W sodium lamp is ($\lambda_{\text{photon}} = 5893\text{\AA}$).

- (1) 7.4×10^{20}
(2) 4.46×10^{22}
(3) 2.7×10^{24}
(4) 5.4×10^{23}

Q100 The major organic product formed from the following reaction is:



- (1) $\text{Ph}-\text{CH}=\text{CH}-\text{CH}=\text{CH}_2$
(2) $\text{Ph}-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_2-\text{NH}_2$
(3) $\text{Ph}-\text{CH}=\text{CH}-\text{CH}(\text{OH})-\text{CH}_3$
(4) $\text{Ph}-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_2-\text{NH}_2$

Q101

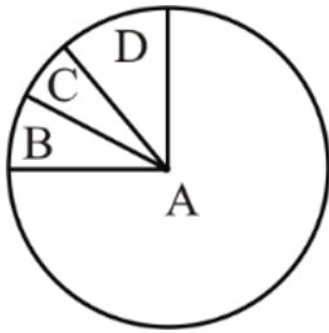
SECTION - A (BOTANY)

All are examples of recent extinction **except**:

- (1) Steller's sea cow and thylacine.
(2) Dodo, Javan tiger and quagga.
(3) Quagga, thylacine and Caspian tiger.
(4) Bali tiger and pigeon.

Q102 Given below is the representation of the extent of global diversity of invertebrates. What groups the four portions (A-D) represent respectively?





- (1) A-Insects, B-Coelenterates, C-Arthropods, D-Molluscs
 (2) A-Annelids, B-Insects, C-Molluscs, D-Porifera
 (3) A-Insects, B-Molluscs, C-Crustaceans, D-Other animal groups
 (4) A-Molluscs, B-Arthropods, C-Coelenterates, D-Insects

Q103 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Interaction of biotic and abiotic components is characteristic for each type of ecosystem.

Reason R: Identification and enumeration of species of an ecosystem gives its species composition.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

Q104 Identify from the given name of scientist who postulated the presence of an adapter molecule.

- (1) Francis Crick
 (2) Friedrich Meischer
 (3) Maurice Wilkins
 (4) Marshall Nirenberg

Q105 Select the **correct** option.

- (1) If detritus is rich in lignin and chitin decomposition rate is fast.
 (2) Aerobiosis inhibit decomposition.
 (3) Moist environment favour decomposition.
 (4) Low temperature favour decomposition.

Q106 Select the **correct** option which shows examples of commensalism.

- (A) Sea anemone and clown fish
 (B) Orchid on mango branch
 (C) Cuckoo and crow
 (D) Barnacles growing on the back of a whale

Choose the most appropriate answer from the options given below:

- (1) A and B only
 (2) A, B and D only
 (3) C and D only
 (4) D only

Q107 Given below are two statements:

Statement I: The Western Ghats have a greater amphibian species diversity than the Eastern Ghats.

Statement II: Scandinavian country like Norway has a greater ecosystem diversity than India.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect
 (2) Statement-I is incorrect and Statement-II is correct
 (3) Both Statement-I and Statement-II are correct
 (4) Both Statement-I and Statement-II are incorrect

Q108 Match List-I with List-II:

List-I		List-II	
(A)	Endemism	(I)	Khasi and Jaintia hills
(B)	Hotspot of India	(II)	Species found in a particular area only
(C)	Sacred grove	(III)	Advanced <i>ex-situ</i> conservation
(D)	Cryopreservation	(IV)	Western Ghats

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
 (2) A-IV, B-III, C-II, D-I
 (3) A-III, B-IV, C-II, D-I
 (4) A-II, B-IV, C-I, D-III



Q109 In charged tRNA amino acid is present at:

- (1) 5' position.
- (2) 3' position.
- (3) anticodon site.
- (4) variable arm.

Q110 Identify the **incorrect** option.

- (1) The food chain or web is formed because of the interdependency.
- (2) Death of organism is the beginning of the detritus food chain/web.
- (3) Primary consumers are carnivores.
- (4) Phytoplankton are producers in an aquatic ecosystem.

Q111 Which of the following statements related to stratification are **incorrect**?

- (A) Horizontal distribution of different species.
- (B) Vertical distribution of different species.
- (C) Species occupy different levels.
- (D) Species occupy same levels.

Choose the most appropriate answer from the options given below:

- (1) A and B only
- (2) A, B and D only
- (3) B and C only
- (4) A and D only

Q112 Which of the following statement is **incorrect**.

- (1) Predators in nature are 'prudent'.
- (2) Some species of insects are cryptically-coloured to avoid being detected easily by the predator.
- (3) Some species of frogs camouflaged to get detected easily by the predator.
- (4) For plants, herbivores are the predators.

Q113 Match List-I with List-II:

List-I (In humans)		List-II (Number)	
(A)	Total number of gene(estimated)	(I)	2968 genes
(B)	Average gene consists of	(II)	3000 bases
(C)	Chromosome 1	(III)	231 genes
(D)	Chromosome Y	(IV)	30,000 genes

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-IV, B-II, C-I, D-III
- (3) A-III, B-IV, C-II, D-I
- (4) A-II, B-I, C-III, D-IV

Q114 _____ secrete digestive enzymes that breakdown dead and waste materials into simple, inorganic materials, which are subsequently absorbed by them.

- (1) Herbivores
- (2) Decomposers
- (3) Carnivores
- (4) Omnivores

Q115 Which is **true** about the structure of heterochromatin?

- (1) Loosely packed; Stain light
- (2) Loosely packed; Stain dark
- (3) Densely packed; Stain light
- (4) Densely packed; Stain dark

Q116 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: In an aquatic ecosystem, the detritus food chain is the major conduit for energy flow.

Reason R: Detritus food chain may be connected with the grazing food chain at some levels.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

Q117 tRNA in bacteria is synthesized by:

- (1) RNA polymerase I.
- (2) RNA polymerase II.
- (3) RNA polymerase III.
- (4) RNA polymerase.

Q118 Identify the **incorrect** statement.

- (1) Crop fields are considered as ecosystem.
- (2) Ecosystem is always of same size.
- (3) Ecosystem is divided into two basic categories, namely the terrestrial and the aquatic.
- (4) Aquarium may also be considered as man-made ecosystems.

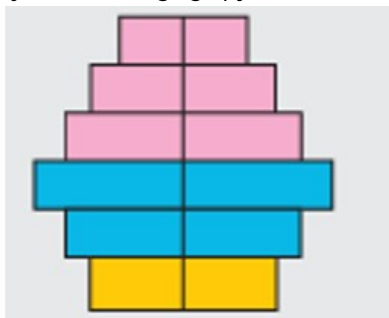
Q119 A nucleotide has the following components:

- (1) two phosphate groups, nitrogenous base, a hexose sugar.



- (2) a phosphorous base, nitrogenous group, a ribose sugar
 (3) a nitrogenous base, phosphate group, a ribose sugar
 (4) two nitrogenous bases, phosphate group, a hexose sugar

Q120 What type of human population is represented by the following age pyramid?



- (1) Stable population
 (2) Declining population
 (3) Expanding population
 (4) Vanishing population

Q121 Select the **incorrect** statement/s.

- A. Each trophic level has a certain mass of living material at a particular time called as the standing crop.
 B. The biomass of a species is expressed in terms of fresh weight.
 C. Measurement of biomass in terms of dry weight is more accurate.
 D. The amount of energy increases at successive trophic levels.
 E. Organisms at each trophic level depend on those at the lower trophic level for their energy demands.

Choose the most appropriate answer from the options given below:

- (1) A and B only
 (2) A, B and D only
 (3) C, D and E only
 (4) D only

Q122 The total number of biosphere reserve and sanctuaries in India are:

- (1) 82 biosphere reserve and 41 sanctuaries.
 (2) 13 biosphere reserve and 421 sanctuaries.
 (3) 85 biosphere reserve and 40 sanctuaries.
 (4) 14 biosphere reserve and 448 sanctuaries.

Q123 Match **List-I** with **List-II**:

List-I		List-II	
(A)	AUG	(I)	Stop codon
(B)	UAA	(II)	Methionine
(C)	UUU	(III)	Phenylalanine
(D)	UAC	(IV)	Tyrosine

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
 (2) A-IV, B-III, C-II, D-I
 (3) A-III, B-IV, C-II, D-I
 (4) A-II, B-I, C-III, D-IV

Q124 What is the approximate annual net primary productivity of whole biosphere?

- (1) 170 billion tons (dry weight) of organic matter.
 (2) 175 million tons (dry weight) of organic matter.
 (3) 180 billion tons (dry weight) of organic matter.
 (4) 155 million tons (dry weight) of organic matter.

Q125 Gause's principle is correlated to:

- (1) commensalism.
 (2) competitive exclusion.
 (3) resource partitioning.
 (4) ammensalism.

Q126 Select the **correct** match.

- A. Aravalli Hills - Rajasthan
 B. Western Ghat regions - Meghalaya
 C. Chanda and Bastar areas - Madhya Pradesh
 D. Sarguja - Gujarat

Choose the most appropriate answer from the options given below:

- (1) A and B only
 (2) A, B and D only
 (3) A and C only
 (4) D only

Q127 _____ is the number of individuals of the population who left the habitat and gone elsewhere during the time period under consideration.

- (1) Immigration (2) Emigration
 (3) Natality (4) Mortality

Q128 _____ is present in DNA only and _____ is present in RNA only.

- (1) Cytosine, thymine
 (2) Thymine, uracil



- (3) Uracil, thymine
(4) Thymine, cytosine

Q129 The formula for logistic growth is:

- (1) $dN / dt = rN$.
(2) $rN / dN = dt$.
(3) $dN / dt = rN \left(\frac{K-N}{K} \right)$.
(4) $dN / dt = rN \left[\frac{N-K}{N} \right]$.

Q130 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: DNA are separated from bulk genomic DNA as different peaks during density gradient centrifugation.

Reason R: Large peaks are referred to as satellite DNA.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q131 Select the **correct** equation regarding GPP.

- (1) $GPP - R = NPP$
(2) $GPP + R = NPP$
(3) $GPP/R = NPP$
(4) All of these

Q132 Pollination in *Ophrys* (a Mediterranean orchid) occurs by:

- (1) insect through copulation mechanism.
(2) bird through pseudocopulation mechanism.
(3) insect through pseudocopulation mechanism.
(4) bird through copulation mechanism.

Q133 Given below are two statements:

Statement I: Of the incident solar radiation more than 50 per cent of it is photosynthetically active radiation (PAR).

Statement II: For deep sea hydro-thermal ecosystem, sun is the only source of energy.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect
(2) Statement-I is incorrect and Statement-II is correct
(3) Both Statement-I and Statement-II are correct
(4) Both Statement-I and Statement-II are incorrect

Q134 If S is species richness, A is area, Z is slope of the line, and the C is Y-intercept, then the species richness will be shown as:

- (1) $S = C + Az$. (2) $S = C + AZ$.
(3) $S = C \cdot Az$. (4) $S = CA^Z$.

Q135 In the Alfred Hershey and Martha Chase experiments the following isotopes were used:

- (1) ^{32}S , ^{30}P (2) ^{31}S , ^{29}P
(3) ^{35}S , ^{32}P (4) ^{34}S , ^{33}P

Q136

SECTION - B (BOTANY)

Match **List-I** with **List-II**:

List-I		List-II	
(A)	Mango	(I)	More than 1.5 million species described (plants and animals)
(B)	Rice	(II)	1,000 varieties in India
(C)	International Union for Conservation of Nature and Natural Resources	(III)	50,000 varieties in India
(D)	Global species diversity	(IV)	7 million

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-I, C-III, D-IV

Q137 The _____ are associations between fungi and the roots of higher plants.

- (1) orchids (2) mycorrhizae
(3) lichen (4) barnacle

Q138 Given below are two statements:

Statement I: The historic Convention on Biological Diversity ('The Earth Summit') was held in Rio de Janeiro.



Statement II: The World Summit on Sustainable Development was held in 2002 in Johannesburg, South Africa.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect
- (2) Statement-I is incorrect and Statement-II is correct
- (3) Both Statement-I and Statement-II are correct
- (4) Both Statement-I and Statement-II are incorrect

Q139 Which of the following is/are produced by plant as defences against grazers and browsers:

- A. nicotine.
- B. caffeine.
- C. quinine.
- D. strychnine.

Choose the most appropriate answer from the options given below:

- (1) A and B only
- (2) A, B, C and D
- (3) A, B and D only
- (4) D only

Q140 Select the **correct** option for **A** and **B**.

In a field experiment, when all the starfish were removed from an enclosed intertidal area more than _____ (**A**) _____ species of invertebrates become extinct within a year, because of _____ (**B**) _____ competition.

- (1) A – twelve , B – intraspecific
- (2) A – ten, B – interspecific
- (3) A – twenty , B – intraspecific
- (4) A – fifteen , B – interspecific

Q141 Read the following statements and select the **correct** option.

- (1) Organisms occupy a specific place in the food chain that is known as their trophic level.
- (2) Plants capture only 50 per cent of the PAR.
- (3) Multidirectional flow of energy occur from the sun to consumers.
- (4) By the process of leaching, water soluble organic nutrients go down into the soil horizon and get precipitated as unavailable salts.

Q142 On the rocky sea coasts of Scotland, the larger and competitively superior barnacle *Balanus* dominates the intertidal area, and excludes the smaller barnacle _____ from that zone.

- (1) *Acacia*
- (2) *Chathamalus*
- (3) oysters.
- (4) trematode parasite.

Q143 Once covering more than 14 per cent of the earth's land surface, these Amazon rain forests now cover no more than _____ per cent.

- (1) one. (2) six.
- (3) ten. (4) twelve.

Q144 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The Monarch butterfly is highly distasteful to its predator (bird) because of a special chemical present in its body.

Reason R: The Monarch butterfly acquires the poisonous chemical during its adult stage by feeding on a poisonous weed.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q145 Bioprospecting deals with exploring _____ level diversity for products of economic importance.

- (1) molecular (2) genetic
- (3) species (4) All of these

Q146 The tRNA pairs through _____ with codes on mRNA through its anticodons.

- (1) peptide bonding
- (2) H-bonding
- (3) 5'-3' phosphodiester bonding.
- (4) 3'-5' phosphodiester bonding.

Q147 Read the following statements and select the **correct** option.

The loss of biodiversity in a region may lead to:

- A. Decline in plant production.
- B. Lowered resistance to drought.
- C. Increased variability in plant productivity.
- D. Increased variability in water use.



E. Increased variability in pest and disease cycles.
Choose the most appropriate answer from the options given below:

- (1) A and B only
- (2) A, B and D only
- (3) C, D and E only
- (4) A, B, C, D and E

Q148 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The *lac* operon consists of one regulatory gene.

Reason R: The *i* gene codes for the inducer of the *lac* operon.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q149 Obtaining industrial products like tannins, lubricants, dyes, resins, perfumes come under:

- (1) broadly utilitarian argument.
- (2) narrowly utilitarian argument.
- (3) ethical argument.
- (4) All of these

Q150 Given below are two statements:

Statement I: The DNA in nucleoid is organised in large loops held by carbohydrates in *E. coli*.

Statement II: Nucleoid is present in *E. coli*.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect
- (2) Statement-I is incorrect and Statement-II is correct
- (3) Both Statement-I and Statement-II are correct
- (4) Both Statement-I and Statement-II are incorrect

Q151 SECTION - A (ZOOLOGY)

The appearance of dry, scaly lesions on various body parts, is seen in which of the following diseases?

- (1) Filariasis

- (2) Dysentery
- (3) Ringworm infection
- (4) AIDS

Q152 Which of the following statements are **correct** about cancer?

A. The chemical carcinogen present in tobacco smoke is identified as a major cause of lung cancer.

B. Cellular oncogenes have been identified in normal cells and activated under certain conditions, which could lead to oncogenic transformation of the cell.

C. CT scans and MRI are beneficial in detecting cancer of the internal organs.

D. Antibodies against cancer-specific antigens are also used for the detection of certain cancers.

- (1) A and B only
- (2) A and C only
- (3) A, B and C only
- (4) A, B, C and D

Q153 HIV decreases the natural immunity of the body by destroying:

- (1) erythrocytes. (2) monocytes.
- (3) T-lymphocytes. (4) B-lymphocytes.

Q154 The use of bio resources by multinational companies and other organisations, without proper authorisations from the countries and people concerned without compensatory payment, is called:

- (1) biopiracy. (2) bioterror.
- (3) bioweapon. (4) biopatent.

Q155 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The recombinant DNA molecule cannot be created, unless one cuts the vector and source DNA with same restriction enzyme.

Reason R: When cut by same restriction enzyme both will have same sticky ends, that can be easily joined by DNA ligase.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.



- (4) Both A and R are true but R is not the correct explanation of A.

Q156 Match List-I with List-II:

List-I			List-II
(A)	Typhoid	(I)	Pathogens infect alveoli
(B)	Pneumonia	(II)	Pathogens usually enter the large intestine
(C)	Common cold	(III)	Pathogens usually enter the small intestine
(D)	Amoebiasis	(IV)	Pathogens infect the nose and respiratory passage only

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-III, B-I, C-IV, D-II
- (3) A-IV, B-III, C-I, D-II
- (4) A-II, B-I, C-IV, D-III

Q157 Which of the following statements is **incorrect** w.r.t primers?

- (1) It is a small chemically synthesised oligonucleotide.
- (2) It is complementary to the template DNA.
- (3) It is a small chemically synthesised polynucleotide.
- (4) Both (1) and (2)

Q158 Select the **incorrect** statement.

- (1) Foreign DNA can be ligated at the *Bam*HI site of ampicillin resistance gene in the vector pBR322.
- (2) The normal *E. coli* do not carry resistance against any of the antibiotics.
- (3) T-DNA transform normal plant cell into a tumor.
- (4) C peptide is not present in the structure of mature human insulin.

Q159 Given below are two statements:

Statement I: Restriction endonucleases are used in genetic engineering to form 'recombinant' molecules of DNA, which are composed of DNA from different sources/genomes.

Statement II: DNA is a hydrophilic molecule.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q160 Read the following statements and choose the option with **correct** sequence for replication of retrovirus.

- A. Production of viral DNA by reverse transcription.
- B. Introduction of viral RNA.
- C. Incorporation of viral DNA in host genome.
- D. New viral RNA produced by the infected cells.

- (1) A→B→C→D
- (2) D→B→C→A
- (3) B→A→C→D
- (4) B→C→A→D

Q161 How many varieties of rice have been estimated to be present in India?

- (1) 200
- (2) 20,000
- (3) 200,000
- (4) 2,000,000

Q162 Select the **correct** statement.

- (1) Morphine is often given to a person who has undergone surgery as a pain killer.
- (2) Chewing tobacco lowers blood pressure and heart rate.
- (3) Cocaine is given to patients after surgery as it stimulates recovery.
- (4) Barbiturates cause insomnia.

Q163 Which of the following diseases is known to be caused by excessive alcohol consumption?

- (1) Liver cirrhosis
- (2) Diabetes mellitus
- (3) Asthma
- (4) Typhoid

Q164 Identify the **correct** statement w.r.t gel electrophoresis.

- (1) DNA is loaded on gel at the anode terminal.
- (2) The most commonly used matrix used in gel electrophoresis is glucose which is a natural polymer extracted from sea weeds.



- (3) Smaller the size of DNA fragment large is the distance it travels through it.
- (4) Pure DNA can be visualised directly by exposing UV radiations.

Q165 Significance of heat shock method in bacterial transformation is to facilitate:

- (1) binding of DNA to the cell wall.
- (2) uptake of DNA through membrane transport proteins.
- (3) uptake of DNA through transient pores in the bacterial cell wall.
- (4) expression of antibiotic resistance gene.

Q166 How can an alien DNA replicate and multiply within the host organism?

- (1) By integrating into the host's chromosome that has an origin of replication.
- (2) By remaining free-floating in the host's cytoplasm.
- (3) By being degraded by the host's enzymes.
- (4) By being targeted for destruction by the host's immune system.

Q167 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Bt cotton contains a gene from *Bacillus thuringiensis* that protects it from insect pests.

Reason R: The Bt toxin gene produces an insecticidal protein that kills insects upon ingestion by disrupting their gut cells.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q168 Read the following statements (A-E).

A. A functional ADA cDNA is introduced into lymphocytes by using retroviral vector and are subsequently returned to the patient.

B. *Bacillus thuringiensis* produce proteins that kill lepidopterans.

C. The source of complementary RNA in RNA interference could be from an infection by

viruses having RNA genome that replicate via an RNA intermediate.

D. In micro-injection method, recombinant DNA is directly injected into the nucleus of an animal cell.

E. The proteins encoded by the *cryIAb* control the cotton bollworms.

How many of the above statements is/are **correct**?

- (1) Two (2) One
- (3) Four (4) Three

Q169 Given below are two statements:

Statement I: Recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase and it results into insertional inactivation of the β -galactosidase gene.

Statement II: At present, about 30 recombinant therapeutics have been approved for human-use the world over.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q170 Which of the following side effects of anabolic steroids is **not** common between males and females?

- (1) Increased aggressiveness
- (2) Mood swings
- (3) Decreased sperm production
- (4) Depression

Q171 Which of the following is a characteristic of cancer cells?

- (1) They grow in a controlled manner.
- (2) They do not invade surrounding tissues.
- (3) They have abnormal growth and division.
- (4) They do not form tumours.

Q172 Palindromic sequence recognised by *EcoRI* is:

- (1) 5' - GAATTC - 3'
3' - CTTAAG - 5'
- (2) 5' - GGAACC - 3'
3' - CCTTGG - 5'



- (3) 5' - CTTAAG - 3'
3' - GAATTC - 5'
(4) 5' - GGATCC - 3'
3' - CCTAGG - 5'

Q173 Given below are two statements:

Statement I: Plague is a bacterial disease.

Statement II: Rheumatoid arthritis is an auto-immune disease.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q174 If we ligate a foreign DNA at *PvuI* site of pBR322, the recombinant plasmid will lose resistance mainly for:

- (1) ampicillin.
(2) kanamycin.
(3) chloramphenicol.
(4) tetracycline.

Q175 Choose the **correct** matching.

(1)	Cellulase	breaks down cellulose.
(2)	Polymerases	break the DNA into fragments.
(3)	Nuclease	separate the two strands of DNA.
(4)	Lysozyme	breaks down the bacterial cytoplasm.

Q176 Find the **incorrect** match.

- (1) Pneumonia- *Streptococcus pneumoniae*
(2) Common cold - Retrovirus
(3) Filariasis- *Wuchereria bancrofti*
(4) Amoebiasis - *Entamoeba histolytica*

Q177 Given below are two statements:

Statement I: RNA interference takes place in all eukaryotic organisms and that involves silencing of specific mRNA.

Statement II: In mammals, insulin is synthesised as a pro-hormone that needs to be processed before it becomes a fully mature and functional hormone.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q178 What is the primary purpose of the Genetic Engineering Approval Committee (GEAC) in India?

- (1) To grant only patents for genetically modified organisms.
(2) To regulate genetic modification research and assess the safety of GM organisms.
(3) To promote the use of traditional farming methods.
(4) To ban all forms of genetic engineering.

Q179 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Smoking causes a deficiency of oxygen in the body.

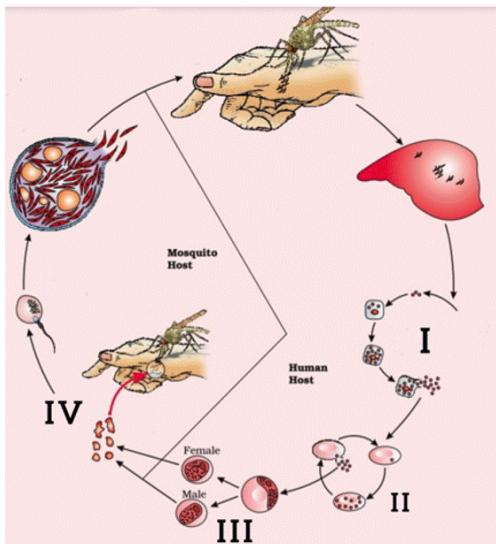
Reason R: Smoking increases carbon monoxide (CO) content in blood and reduces the concentration of haembound oxygen.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q180 Given below diagram is of life cycle of *Plasmodium*. Identify the stage (s) in which the parasite reproduces asexually.





- (1) I only
(2) IV only
(3) I and IV only
(4) I and II only

Q181 Read the following statements (A-E).

- A. Peyer's patches are secondary lymphoid organs.
B. Only the bone marrow is involved in the development and maturation of lymphocytes.
C. IgA antibodies are associated with allergic responses.
D. The receptors for cannabinoids are principally present in the gastrointestinal tract.
E. The drugs can be obtained from fungi.

Choose the option with **incorrect** statements.

- (1) A, B and D only
(2) B and D only
(3) B, D and E only
(4) B, C and D only

Q182 Which of the following statements is **incorrect**?

- (1) Dysentery and diphtheria are bacterial diseases in man.
(2) Vaccination involves the introduction of either inactivated or weakened pathogens into the body.
(3) The antibodies produced in response to vaccination do not provide long-term immunity.
(4) AIDS was first reported in 1981 and it has spread all over the world killing more than 25 million persons.

Q183 Given below are two statements:

Statement I: Nematode-specific genes were introduced into the host plant by using

Agrobacterium vector.

Statement II: A single stranded DNA or RNA, tagged with a radioactive molecule (probe) is allowed to hybridise to its complementary DNA in a clone of cells followed by detection using autoradiography.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q184 Why is traditional knowledge related to bio-resources important for developing countries?

- (1) It is only beneficial for industrialised nations.
(2) It can be exploited to develop modern applications and save costs in commercialisation.
(3) It is of no value in scientific research.
(4) It leads to decreased biodiversity.

Q185 Loss of the property of the body to recognise and differentiate between self-cells and non-self cells, may lead to:

- (1) graft rejection.
(2) passive immunity.
(3) auto-immune disease.
(4) allergic response.

Q186

SECTION - B (ZOOLOGY)

If a recombinant DNA bearing gene for ampicillin resistance is transferred successfully intact into *E. coli* cells and the host cells, are spread on agar plates containing ampicillin, then:

- (1) both transformed and non-transformed recipient cells will die.
(2) both transformed and non-transformed recipient cell will grow.
(3) transformed recipient cells will grow and non-transformed recipient cells will die.
(4) transformed recipient cells will die and non-transformed recipient cells will grow.

Q187 Which of the following neurotransmitters transport is interfered by cocaine?

- (1) Serotonin
(2) Dopamine
(3) Norepinephrine
(4) GABA



Q188 'rop' in pBR322 codes for:

- (1) proteins involved in the replication.
- (2) tetracycline resistance protein.
- (3) ampicillin resistance protein.
- (4) sequence from where replication starts.

Q189 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The downstream processing and quality control testing are the same for every product.

Reason R: Each product has different properties and requirements for its final formulation and market readiness.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q190 Mucosa associated lymphoid tissue (MALT) constitute about _____ per cent of lymphoid tissue in human body.

Choose the option which fills the blank **correctly**.

- (1) 40
- (2) 50
- (3) 60
- (4) 70

Q191 Which of the following initiatives is part of the Indian Patents Bill amendment?

- (1) Giving inadequate compensations.
- (2) Research and development initiatives.
- (3) Promote unauthorised exploitation of bio-resources.
- (4) Giving unequal benefit sharing.

Q192 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Metastasis is the most feared property of malignant tumours.

Reason R: Cells sloughed from malignant tumours reach distant sites through blood, and spread a new tumour at other sites of the body.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is not the correct explanation of A.

Q193 What is the primary goal of inserting alien DNA into a cloning vector and transferring it into a host cell?

- (1) To enhance the genetic diversity of the host organism.
- (2) To produce a small number of copies of the alien DNA.
- (3) To express and produce a desirable protein.
- (4) To disrupt the host cell's DNA.

Q194 Match List-I with List-II:

List-I		List-II	
(A)	Emphysema	(I)	Based on antigen-antibody interaction.
(B)	Rosie	(II)	Can be treated by α -1 antitrypsin
(C)	ELISA	(III)	Protein enriched milk
(D)	PCR	(IV)	Amplification of DNA.

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-I, B-III, C-IV, D-II
- (3) A-I, B-II, C-III, D-IV
- (4) A-IV, B-III, C-II, D-I

Q195 Sustained high fever (39° to 40°C) weakness, stomach pain, constipation, headache, loss of appetite and intestinal perforations in severe cases are symptoms of:

- (1) typhoid caused by the virus *Salmonella typhi*.
- (2) typhoid caused by bacterium *Salmonella typhi*.
- (3) pneumonia caused by the bacterium *Streptococcus pneumoniae*.
- (4) malaria caused by protozoan *Plasmodium*.

Q196 Read the following statements (A-D).

A. Restriction enzymes belong to a large class of enzymes called nuclease.

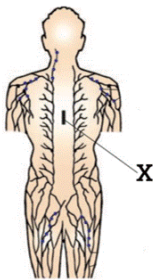


- B. *Hind II* cuts DNA at a specific sequence of six base pairs.
- C. Each restriction endonuclease recognises a random nucleotide sequences in the DNA.
- D. Restriction enzymes cut the sugar-phosphate backbone of DNA.

Which of the above given statements are **correct** for restriction enzymes?

- (1) A, B and D only
- (2) A, B and C only
- (3) A, C and D only
- (4) A and D only

Q197 Identify 'X' and choose the **correct** option in the given diagram.



- (1) The bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced.
- (2) Lymph nodes serve to trap the micro-organisms or other antigens, which happen to get into the lymph and tissue fluid.
- (3) The spleen mainly contains lymphocytes and phagocytes and acts as a filter of the blood by trapping blood-borne micro-organisms.
- (4) The thymus is a lobed organ located near the heart and beneath the breastbone and quite large at the time of birth but keeps reducing in size with age.

Q198 Given below are two statements:

Statement I: The foetus receives some antibodies from their mother, through the placenta during pregnancy and provides passive immunity to the foetus.

Statement II: Active immunity is slow and takes time to give its full effective response.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q199 Match **List-I** with **List-II**:

List-I			List-II
(A)	Physiological barrier	(I)	Tears from eyes
(B)	Physical barrier	(II)	Interferons
(C)	Cellular barrier	(III)	Skin
(D)	Cytokine barrier	(IV)	Macrophages in tissues

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-III, C-IV, D-I
- (4) A-III, B-I, C-II, D-IV

Q200 Read the following statement (**A- E**).

A. Memory B and T cells generated during vaccination that recognise the pathogen quickly on subsequent exposure and overwhelm the invaders with a massive production of antibodies.

B. Immunity simply means 'absence of disease' or 'physical fitness'.

C. Heroin is generally taken by snorting and injection and is a depressant and slows down body functions.

D. The primary immune response is of high intensity.

E. Marijuana, hashish, charas and ganja are generally taken by inhalation and oral ingestion and are known for their effects on cardiovascular system of the body.

Choose the option with the **correct** statements.

- (1) A, C and E only
- (2) B, C and D only
- (3) A, B and C only
- (4) C, D and E only



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